

TASK 1 -REPORT

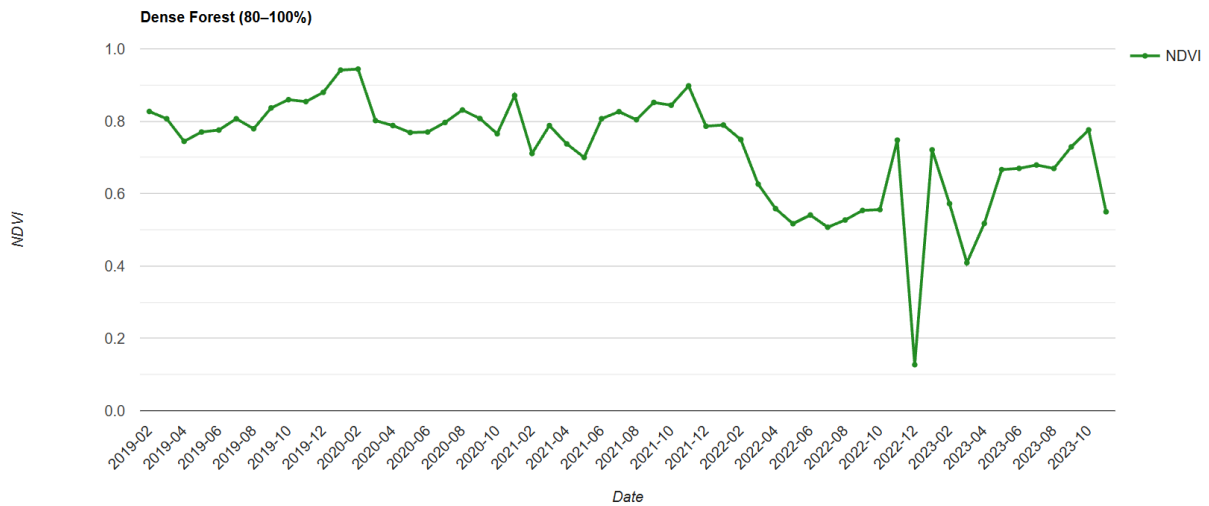
1. Selected Data Points

The following 5 Hansen-confirmed loss pixels within the Bavaria bounding box were selected and analysed across a multi-year window (2019–2023):

Pixels	Category	Tree Cover (%)	Loss Year	Coordinates
Pixel 1	Dense 1	98	2022	(11.4376, 48.0569)
Pixel 2	Dense 2	89	2020	(11.62360, 47.98457)
Pixel 3	Medium	64	2022	(11.29928, 48.04259)
Pixel 4	Sparse	47	2020	(11.448847, 48.015903)
Pixel 5	Very Sparse	21	2020	(11.604945, 47.873129)

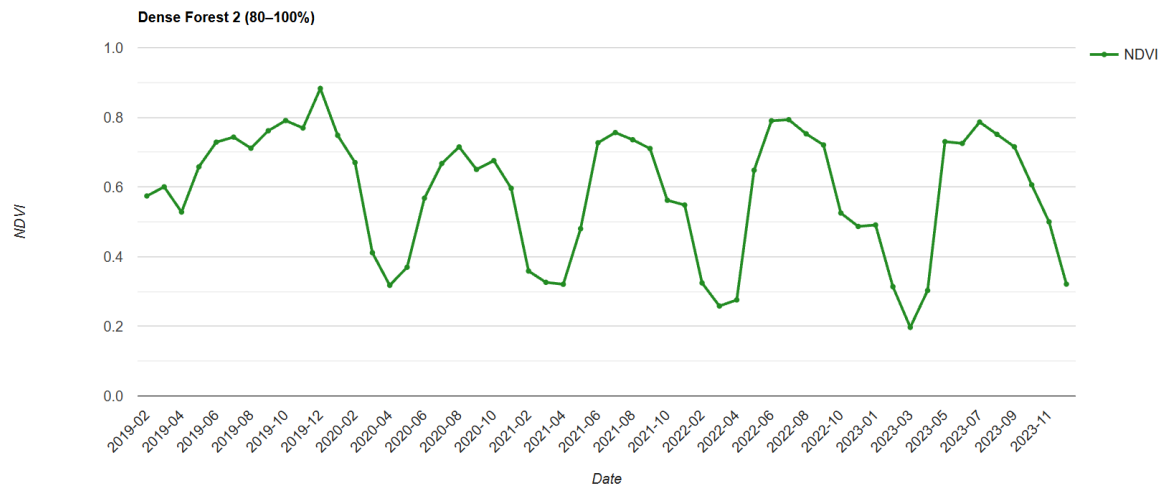
2. NDVI Time-Series Charts:

A) Dense Forest Pixel (1) -



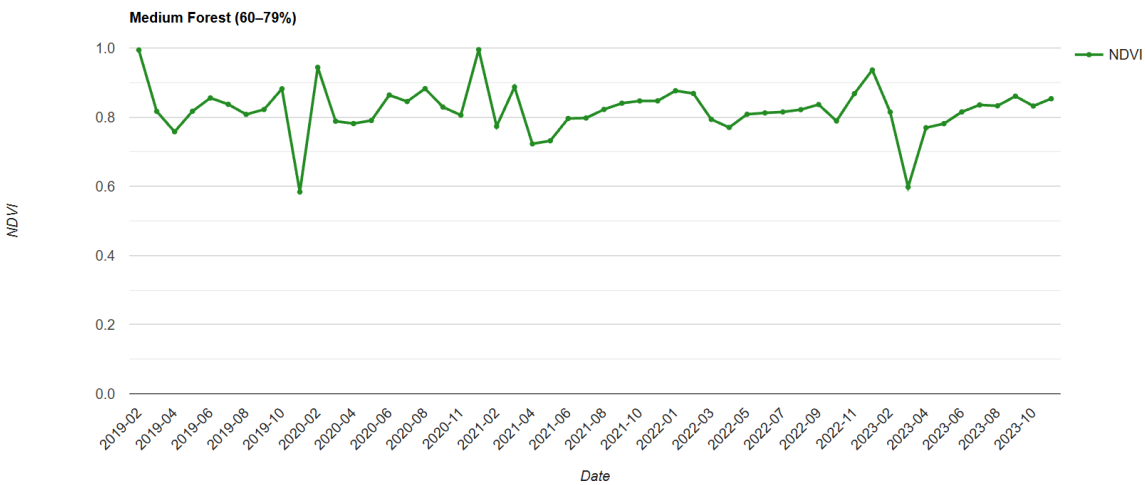
Dense Forest Pixel (1) NDVI Time Series

B) Dense Forest Pixel (2) -



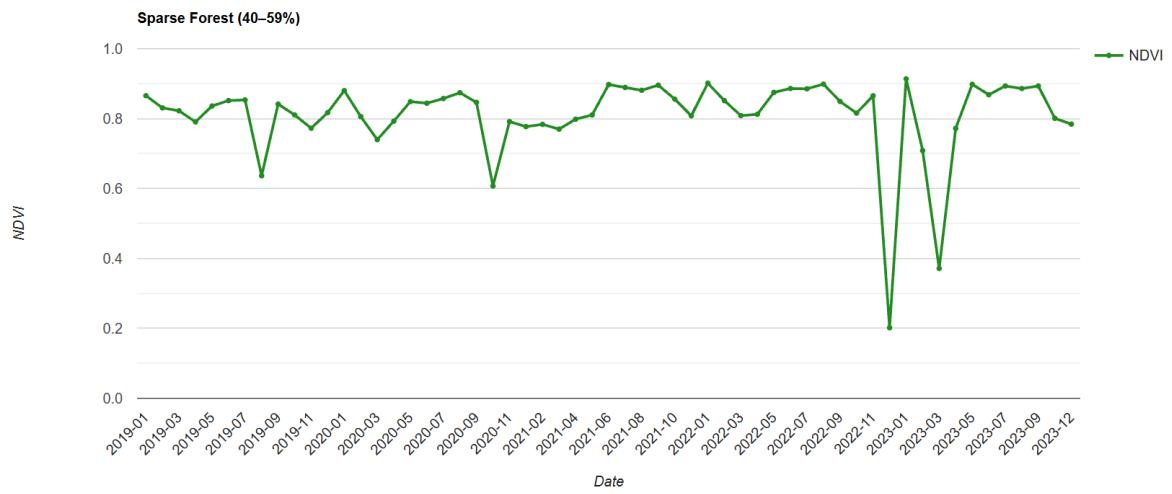
Dense Forest Pixel (2) NDVI Time Series

C) Medium Forest Pixel –



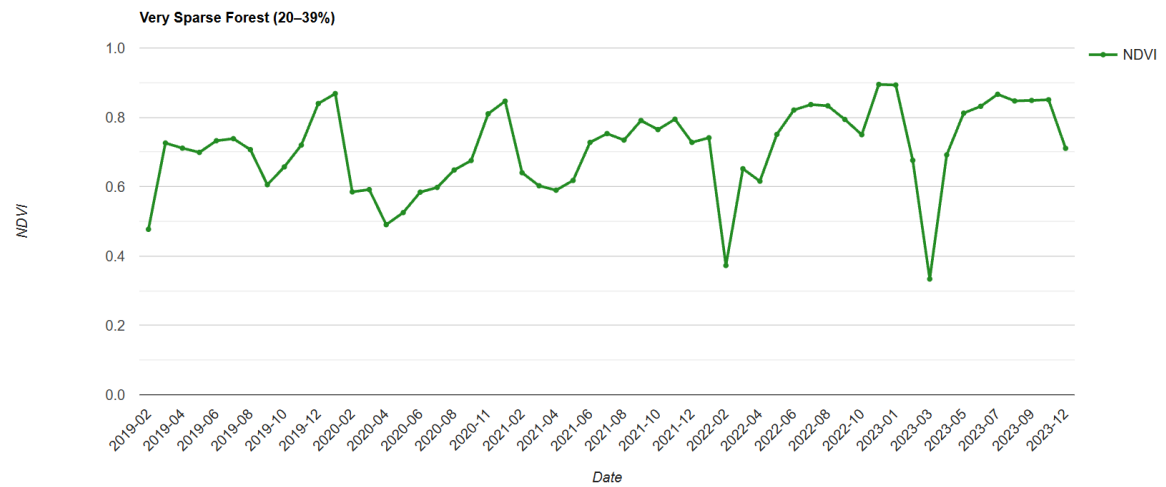
Medium Forest Pixel NDVI Time Series

D) Sparse Forest Pixel -



Sparse Forest Pixel NDVI Time Series

E) Very Sparse Forest Pixel -



Very Sparse Forest Pixel NDVI Time Series

Pixel Category	Baseline Seasonal Drop (Δ NDVI)	Observed Clearing Drop (Δ NDVI)	Clear > Seasonal?
Pixel 1: Dense 1 (98%)	≈ 0.11	≈ 0.30	Yes
Pixel 2: Dense 2 (89%)	≈ 0.43	≈ 0.03	No
Pixel 3: Medium (64%)	≈ 0.12	≈ 0.02	No
Pixel 4: Sparse (47%)	≈ 0.06	≈ 0.00	No
Pixel 5: Very Sparse (21%)	≈ 0.25	≈ 0.00	No

Verdicts per Pixel:

1) Pixel 1 (Dense 1): [LOSS YEAR: 2022]

Results:

- Seasonal NDVI Drop $\approx$ **0.11**
- Clearing Event NDVI Drop $\approx$ **0.30**

Verdict:

The clearing-event drop clearly exceeded the normal seasonal variation, indicating a sustained reduction in vegetation consistent with the Hansen 2022 forest-loss record. The isolated NDVI value of 0.127 in December 2022 was treated as an outlier because of its immediate recovery.

2) Pixel 2 (Dense 2): [LOSS YEAR: 2020]

Results:

- Seasonal Drop: ≈ 0.43
- Clearing Event Drop: ≈ 0.03

Verdict:

The clearing-event NDVI drop is much smaller than the normal seasonal variation. The pixel shows a consistent annual recovery to high summer NDVI values after 2020,

suggesting that the reported Hansen forest loss is not reflected as a sustained canopy reduction at this location and may represent a localized disturbance, rapid vegetation recovery, or a mismatch between the Hansen loss label and the Sentinel-2 time series. [Assumptions based on online

* The seasonal NDVI drop was estimated from the typical annual maximum (summer) and annual minimum (late winter/early spring), representing the natural vegetation cycle.

3) Pixel 3 (Medium): [LOSS YEAR: 2022]

Results:

- Seasonal NDVI Drop $\approx$ **0.12**
- Clearing Event NDVI Drop $\approx$ **0.02**

Verdict:

The clearing-event NDVI drop is considerably smaller than the normal seasonal variation. No persistent decline in canopy greenness is visible after the reported 2022 Hansen loss year. This suggests that this location does not exhibit a clear forest-loss signature in the Sentinel-2 NDVI time series.

* The seasonal NDVI drop was estimated using the typical summer maximum and representative annual minimum from the stable pre-loss period (2019–2021). The anomalous winter values which didn't align with expected forest phenology were not considered.

*This pixel sample is evidence to state that NDVI alone fails to detect some Hansen-confirmed forest loss events.

4) Pixel 4 (Sparse): [LOSS YEAR: 2020]

Results:

- Seasonal NDVI Drop $\approx$ **0.06**
- Clearing Event NDVI Drop $\approx$ **0.00**

Verdict:

The clearing-event drop is smaller than the normal seasonal variation, and no sustained reduction in NDVI is observed after the reported 2020 Hansen loss year. This pixel does not exhibit a clear forest-loss signature in the Sentinel-2 NDVI record, suggesting either a localized disturbance, rapid vegetation recovery, or a mismatch between the Hansen loss label and the NDVI response.

*The seasonal NDVI drop was estimated from the representative summer maximum and annual minimum observed during the stable pre-loss period. Short anomalous drops with instant recovery were masked as outliers.

*This pixel sample is evidence to state that NDVI alone fails to detect some Hansen-confirmed forest loss events.

5) Pixel 5 (Very Sparse): [LOSS YEAR: 2020]

Results:

- Seasonal NDVI Drop $\approx$ **0.25**
- Clearing Event NDVI Drop $\approx$ **0.00**

Verdict:

The clearing-event NDVI drop is substantially smaller than the normal seasonal variation. The NDVI time series does not show a persistent decline following the reported 2020 Hansen loss year, suggesting that the forest-loss signal is not clearly detectable using NDVI alone at this low-canopy site.

*Low NDVI observations were isolated as they seemed to recovered immediately in the subsequent months.

GEE Code Link: <https://code.earthengine.google.com/b00799ae7845253af49bbf6e40fa809e>

GitHub Repository Link: <https://github.com/SiddharthBhimpure/bavaria-forest-change-analysis.git>

END OF REPORT